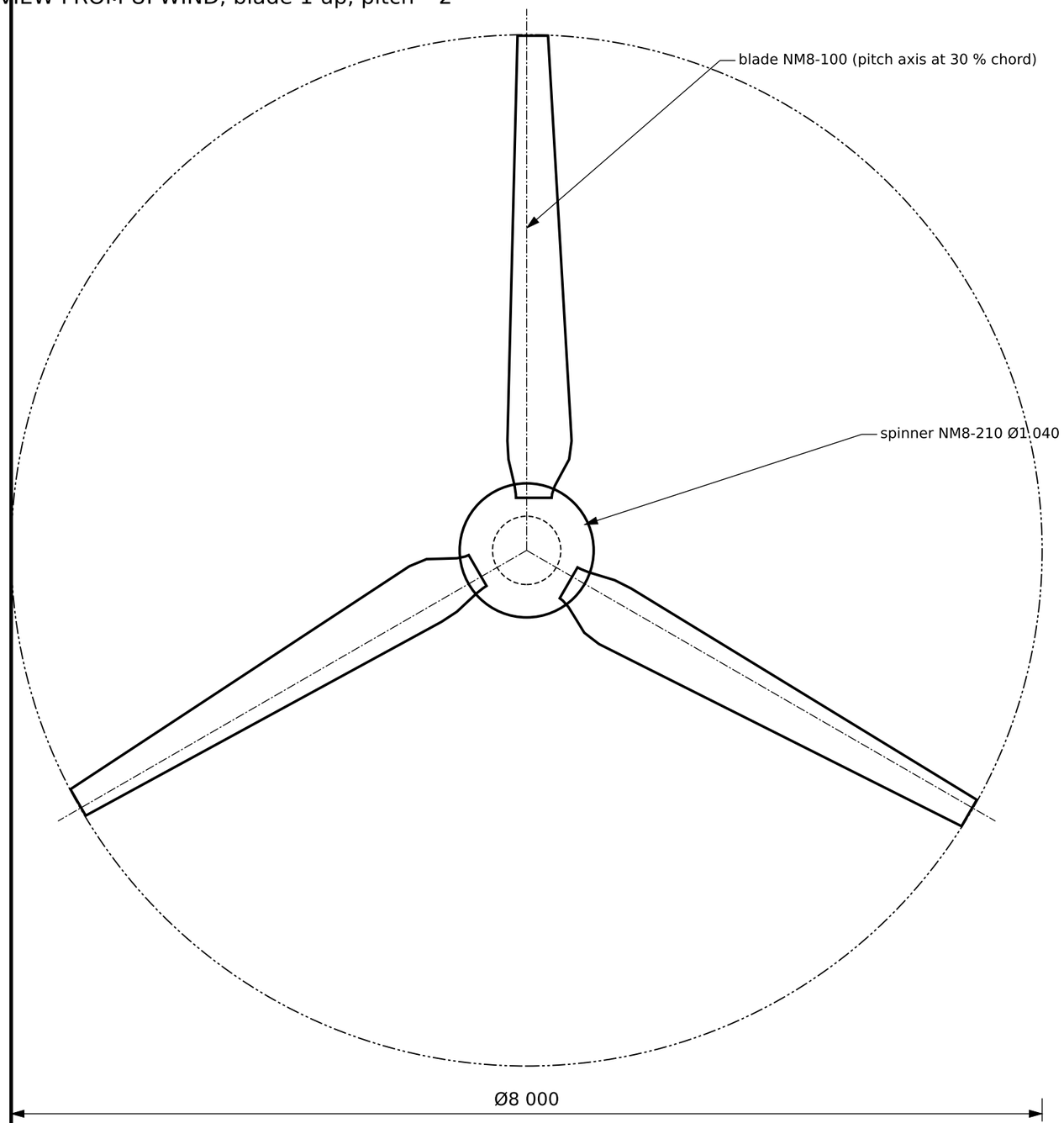
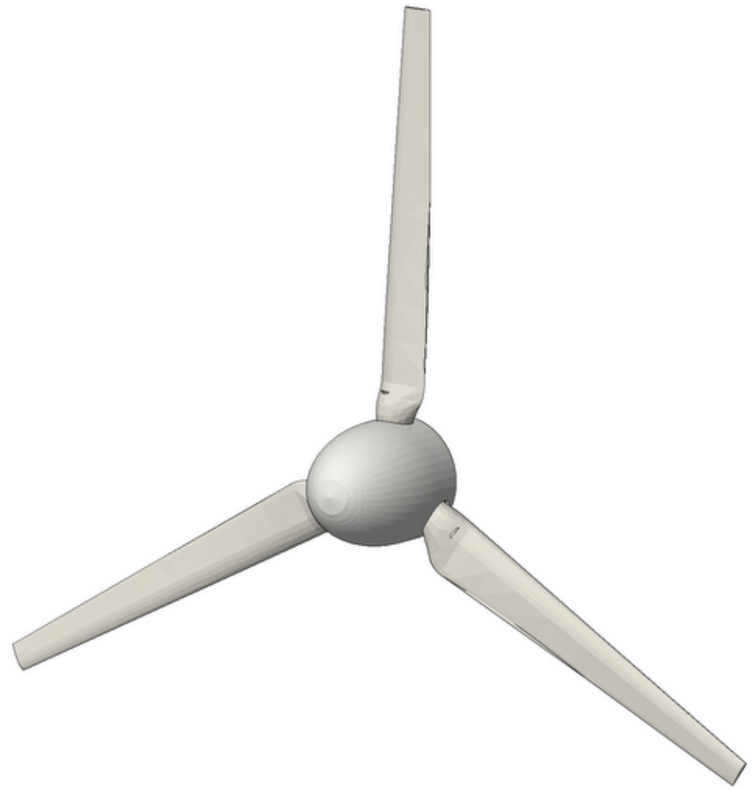
VIEW FROM UPWIND, blade 1 up, pitch −2°



AXONOMETRIC VIEW (not to scale)



Item	Value
Blades	3 × 34.1 kg, carbon UD caps (D3)
Blade root	Ø280, 12 × M16 inserts PCD 245
Pitch bearings	4-point Dpw 300, balls 30, Z 28
Pitch drive	one electric cylinder, crank r140, collective
Hub	S355NL weldment 180 kg (budget)
Rotor inertia	≈ 385 kg m²
Pitch range	−2° (fine) … 92° (feather)

NOTES

1. Assembly order (NM8-020 = build book steps 8–10): hub pre-assembled with bearings, cranks, links, yoke, cylinder and nose frame on the bench; mounted on the drum; blades fitted with the cylinder at the −2° end stop; spinner last.
2. Static-moment matched blade set: moments within 0.5 % (balance by root ballast pocket); mark each blade with its hub position.
3. Bolt torques NM8-900. Blade root bolts M16 10.9: 280 Nm, re-torque after 50 h.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE M8 (ICELAND) · Ø8.0 m, COLLECTIVE PITCH · 8 m TOWER

ROTOR ASSEMBLY · HUB, PITCH SYSTEM, BLADES, SPINNER

DRAWING NO.	REV / DATE	SCALE (A3)
NM8-020	A · 2026-11-06	1:40
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
see BOM	≈ 455 kg	Claude / not checked